Research Question and Initial Hypothesis

Is contrast the single most important factor for text readability, particularly on a digital reading device, or can other, non contrast-maxing, colour combinations (i.e. outside of pure black-on-white or white-on-black) produce equal or better reading performance and lower visual fatigue, based on the reading task, ambient lighting, and environmental factors?

While it is obvious that a minimum amount of contrast is necessary for legibility, contrast alone is not sufficient to predict readability. Outside of just the contrast threshold, other properties of a colour combination, such as display polarity (dark-on-light versus light-on-dark), hue/temperature (warm versus cool backgrounds), screen brightness, and the environmental ambient lighting, will all measurably affect a user's reading speed, error rate, and visual comfort. Specifically, we hypothesize that certain middle-contrast combinations (e.g. dark grey text on a warm off-white peach or cream background) will match or even exceed the reading performance of contrast-maxing black text on white backgrounds, while producing lower reported eye strain during sustained reading.

Significance of the Question and Intended Audience

Nearly every office worker, student, and casual mobile user now spends hours per day reading text on screens, and the default assumption baked into most design has been simple and constant: higher contrast is better. The Web Content Accessibility Guidelines (WCAG 2.1) indicate that for text readability, a minimum luminance contrast ratio (4.5:1 for normal text) must be maintained, and designers have begun to treat black-on-white as the gold standard [^13] in web content. Yet the rapid consumer adoption of "dark mode," "reading modes" that use warmer tones, and yellow tinted "nightlight" overlays marketed to readers with dyslexia suggests that the users themselves do not necessarily experience maximum contrast as the universal optimal method. Recent experiments show that dark mode can reduce visual fatigue under low ambient light when contrast is kept high [^15] and that within dark interfaces, the choice of text colour can change the fatigue a user experience; yellow-on-black produced the lowest visual fatigue while red-on-black produced the most [^3]. If contrast is an incomplete predictor of readability, then accessibility standards, operating system defaults, e-readers, and mobile software may all be leaving measurable performance and comfort gains on the table.

The question concerns people, not just interface design. Studies have shown that warm background colours improved reading performance for readers both with and without dyslexia [^11] and that display polarity affects legibility for both younger and older adults [^10]. Furthermore, studies have found that small, low-contrast text is disproportionately harmful to readers over 40 [^14], causing increase eyestrain at significantly higher rates. A clearer answer could directly improve reading experiences for many users, especially readers of higher ages using digital interface, as well as anyone who reads on a screen for long stretches particularly at night or in dim lighting.

Outside of the intended audience for this proposed research include researchers studying legibility and visual ergonomics, accessibility standards bodies such as the W3C (whose contrast centered guidelines this research directly questions), and UI/UX designers/front-end developers who need evidence-based defaults for text colours, dark-mode palettes, and reading modes. The research can also be used to provide deeper insights on the effects of contrast for those with reading/visual impairments, such as dyslexia, and can answer accessibility questions related to the needs of those users.

Background and Context

Research on text/background colour combinations can be dated back to studies of cathode ray displays. In an early study, Pastoor (1990) evaluated hundreds of foreground/background colour combinations and found that, once contrast was adequate, chromaticity (the actual hues/colours chosen) had little effect on text legibility --- though it did affect a user's subjective preference [^9]. Later work on modern displays largely reinforced the importance of contrast, following the convention that higher luminance or brightness contrast consistently improves legibility more than hue alone, testing across web pages [^5], LCD displays [^6], and negative polarity (dark background) interfaces. Increasing text brightness improved legibility and comfort while changing background hue alone had little effect when brightness was held constant [^8]. This is further emphasized in modern contrast-based standards such as WCAG [^13], where specifics on the minimum recommended contrast ratio are used as the standard for readability.

However, some modern studies and findings complicate the simple "higher contrast is better" theory. For instance, in the case of polarity: black-on-white and white-on-black have identical WCAG contrast ratios (21:1), yet they are not equivalent for readers. Studies using proofreading and screen reading within their methodologies repeatedly find that a "positive polarity advantage", meaning dark text on light backgrounds, outperforms the reverse, irrespective of ambient illumination and colour contrast [^1] for both younger and older adults [^10]. In these studies, the advantage is attributed to smaller pupil size and sharper retinal images under brighter backgrounds. Notably, the effect interacts with context; in "glance-like reading", negative polarity performed worst under dark ambient lighting [^2], while under sustained low-light use, high-contrast dark mode reduced visual fatigue [^15].

Additionally, there are effects that hue has that go beyond contrast. In a brand-icon search task, colour combination mattered even more than luminance contrast overall, with yellow-on-black, yellow-on-blue, and white-on-blue ranking highest [^7]. Specific pairings, such as green-on-black and green-on-white, improved text-recognition efficiency on digital devices, while some blue text combinations underperformed despite adequate contrast [^4]. Studies involving eye-tracking further dissociates attention capture from sustained processing, with yellow backgrounds capturing attention fastest, while other different backgrounds better supported ongoing reading [^12]. Most strikingly for this proposal, Rello and Bigham (2017) found in a study of 341 readers (89 with dyslexia) that warm background colours (peach, orange, yellow) produced significantly faster reading than cool backgrounds for participants with and without dyslexia, despite those backgrounds having lower contrast against black text than pure white does.

Detailed Analysis (empirical paper): Hall & Hanna (2004)

The paper by Hall and Hanna (2004) asks how the text/background colour combination of a web page affects four outcomes: readability, retention of the page's content, perceived aesthetic quality, and behavioural intention (e.g. intention to purchase or to revisit). The motivation behind the study was obvious: web design guidance about colour was largely folklore, and earlier legibility studies used short artificial stimuli (isolated words on CRT monitors) rather than realistic web pages. Because businesses and educators were rapidly moving content online, the authors argue that designers needed evidence connecting colour choices not only to low-level legibility but also to higher-level cognitive and behavioural outcomes.

The article reviews other research articles, showing that the legibility of colour combinations is driven largely by luminance contrast, with studies such as Pastoor (1990) , which had participants rate nouns shown in 792 colour combinations, that found little effect of hue once contrast was controlled. Their paper found that some colour effects found in earlier studies were mediated by other display variables such as font type, putting into question the experiments done prior. They also found that colour reliably influences affective responses such as arousal and preference, of which contributed to their own experiments design and goals. What prior work had not established was whether colour combinations effect deeper processing, such as retention of studied material, or downstream behaviour (such as purchase intention) on realistic web pages, and how readability and aesthetics might trade off against each other.

The paper's contribution is to extend colour combination research from low-level legibility to higher-level outcomes, using realistic use cases, such as educational and commercial web pages, as the basis for testing these stimuli. The paper resulted in some key findings: greater contrast improved readability, colour combination had no significant effect on retention, preferred chromatic colours (e.g., blues) raised aesthetic ratings and purchase intention, and aesthetic ratings were significantly related to behavioural intention. The practical contribution is the insight that the "optimal" colour scheme depends on the material's purpose, using maximum-contrast achromatic schemes for knowledge transfer and non-monochromatic schemes for commercial appeal.

Their research review resulted in a controlled experiment with 136 participants. The primary independent variable was the text/background colour combination, assigned to subjects at four differing levels of contrast ratio and chromaticity. Participants were assigned to one of the four colour-combination conditions as well as page content type. Every participant studied was given both an educational page and a commercial page, so that each participant served as their own control across content types. The dependent variables were readability, retention of page content (measured with test questions after studying), self-rated aesthetic quality, and behavioural intention measured through questionnaires designed for the experiment so that readability and aesthetics would not be incorrectly interpreted in a single preference score. Other confounding variables were attempted to be handled through standardization (same page layouts and content across conditions), although the authors acknowledge that monitor luminance could not be fully controlled, though this was seemingly intentional since they state that real user displays also vary. The design therefore supports causal inference from colour combination to the measured outcomes via random assignment and control of the stimulus materials, while its main threats to validity are the limited number of colour conditions and the uncontrolled display hardware.

Further Hypothesis

A deeper, more specific hypothesis that could be tested is that sustained reading sessions (20+ minutes) on modern LCD/OLED displays under normal indoor lighting, a moderate-contrast, positive-polarity, warm colour combination (e.g., near-black text on a warm off-white background) will produce reading speed and comprehension equal to or greater than maximum contrast black-on-white, as well as significantly lower self-reported visual fatigue. Based on my initial research, we also believe that this advantage have a larger impact for readers with dyslexia than for typical readers, though specific testing and data standardization will be required to adequately determine if this is the case. This hypothesis deliberately puts into question the contrast-first account (which predicts black-on-white should win or tie on all measures) against a multi-factor account which has been supported by the polarity, hue, and fatigue findings reviewed above. We predict that polarity, hue temperature, and background luminance contribute should all independently to readability in unequal parts, all of which could be dependent on the task intended. A natural extension, following Xie et al. (2021) and Dobres et al. (2017), would be to add ambient lighting (bright vs. dark room) as another factor to test whether the optimal combination changes in specific conditions or time of day.

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